SESSION 1 — MORNING

Time Value of Money

The most fundamental concept in all of finance

Session Roadmap — 2 Hours

Six sections, building from first principles to real-world CFA examples. Every concept connects forward.

01 **What Is an Interest Rate?**
3 interpretations • decomposing r • CFA LM1 bond yield build-up

02 **Return Measurement**
Holding period • geometric mean • harmonic mean • money-weighted vs time-weighted

03 **Future Value & Compounding**
FV formula • compounding frequency • EAR

04 **Present Value & Discounting**
PV formula • timelines • CFA India bond • CFA Greece bond

05 **NPV & IRR**
Project evaluation • decision rules • conflict resolution

06 **Annuities, Perpetuities & DDM**
Mortgage • KB Financial perpetual • Shipline two-stage DDM

Learning Objectives

By the end of this session you will be able to:

- **Interpret an interest rate**
  Describe r as required return, discount rate and opportunity cost. Decompose r into its five risk components.
- **Calculate return measures**
  Compute arithmetic, geometric and harmonic means. Distinguish money-weighted from time-weighted return.
- **Calculate FV and PV**
  Apply compounding and discounting formulas. Handle multiple compounding frequencies. Compute EAR.
- **Price bonds using TVM (CFA LM2)**
  Value discount bonds, coupon bonds, semiannual bonds and perpetual bonds using real CFA worked examples.
- **Apply NPV and IRR**
  Evaluate projects. Know when NPV and IRR conflict and which to trust. Understand IRR limitations.
- **Price annuities, perpetuities & DDM**
  Use shortcut formulas for mortgages, KB Financial perpetual bond and Shipline two-stage dividend model.

SECTION 01

What Is an Interest Rate?

Three Ways to Read an Interest Rate (CFA LM1)

If USD 9,500 today and USD 10,000 in one year are equivalent, the interest rate is 5.26%. CFA LM1 uses this to define three interpretations.

$$r = (10{,}000 - 9{,}500) / 9{,}500 = \text{USD } 500 / \text{USD } 9{,}500 = 5.26\%$$

1 Required Rate of Return

The MINIMUM return an investor will accept. Below 5.26%, the investor prefers to hold USD 9,500 today rather than commit it for a year.

2 Discount Rate

The rate used to translate a future value back to a present value. USD 10,000 discounted at 5.26% equals USD 9,500 today.

3 Opportunity Cost

What is foregone by choosing to invest. Spending USD 9,500 now means giving up the 5.26% return that could have been earned.

Decomposing an Interest Rate (CFA LM1)

r = Real Risk-Free + Inflation + Default Risk + Liquidity + Maturity Premium

Real Risk-Free Rate	0.5-1%	Return on a completely risk-free asset with zero inflation. Reflects investor time preference for consumption today over tomorrow.
Inflation Premium	2-3%	Compensates for expected loss of purchasing power. Fisher approximation: Nominal rate = Real rate + Expected inflation.
Default Risk Premium	0-5%+	Compensation for potential non-payment. Corporate bonds yield more than government bonds because of this premium.
Liquidity Premium	0.5-1.5%	Compensation for assets that are hard to sell quickly at fair value. Small-issuer bonds carry higher illiquidity premiums.
Maturity Premium	0.5-2%	Longer bonds are more sensitive to rate changes. Investors demand extra return for locking up capital for longer periods.

Building a Corporate Bond Yield — Step by Step

A 5-year BBB-rated UK corporate bond. Each component adds a specific risk premium to compensate the investor.

Component	Premium Added	Cumulative Yield	Rationale
Real risk-free rate	0.75%	0.75%	Time preference, no risk
+ Inflation premium	2.00%	2.75%	CPI expected 2% p.a.
+ Default risk premium	1.80%	4.55%	BBB-rated issuer
+ Liquidity premium	0.50%	5.05%	Medium-size issuer, infrequent trading
+ Maturity premium	0.45%	5.50%	5-year vs 1-year tenor

CONCLUSION

Required yield = 5.50%. If the bond coupon = 5.50%, it prices at PAR (100).
Coupon above 5.50% → premium bond (price > 100). Coupon below 5.50% → discount bond (price < 100).
This premium decomposition is the foundation of all credit and fixed income analysis.

SECTION 02

Return Measurement (CFA LM1)

Holding Period Return & Arithmetic Mean

HPR = (End Price − Begin Price + Income) / Begin Price

The HPR is the single-period total return. For multiple periods, the arithmetic mean averages them but does NOT give the compound growth rate.

THE TRAP: Arithmetic Mean Overstates Multi-Period Performance

Fund A: Year 1 = +50%, Year 2 = −50%.
Arithmetic mean = (+50 − 50)/2 = 0% — sounds like break-even.
But: USD 1,000 × 1.50 × 0.50 = USD 750. You LOST 25%.
The geometric mean correctly shows: [(1.50)(0.50)]^0.5 − 1 = −13.4% per year.
Rule: Use arithmetic mean for expected FORWARD-looking returns. Use geometric mean for HISTORICAL multi-period compound growth.

Arithmetic mean ≥ Geometric mean always. The gap grows with return volatility.

Geometric Mean — CFA LM1: Fund Y Practice Problem

$$R_g = [(1+R_1)(1+R_2)\ldots(1+R_n)]^{(1/n)} - 1$$

FUND Y — directly from CFA Vol 1, LM1 Practice Problem 3
Five-year annual returns: Year 1 = 19.5%, Year 2 = −1.9%, Year 3 = 19.7%, Year 4 = 35.0%, Year 5 = 5.7%.

Step 1: Convert to (1 + R) factors: 1.195 × 0.981 × 1.197 × 1.350 × 1.057
Step 2: Product = 1.195 × 0.981 × 1.197 × 1.350 × 1.057 = 1.9841 (approx)
Step 3: Take the 5th root: 1.9841^(1/5) = 1.1487
Step 4: Subtract 1: 1.1487 − 1 = 14.87% ≈ 14.9%

CFA answer: Geometric mean = 14.9%

What 14.9% means

USD 1,000 invested at the start grows to USD 1,000 × 1.149^5 = USD 1,984 — identical to the actual portfolio.

Arithmetic mean here

= (19.5 − 1.9 + 19.7 + 35.0 + 5.7)/5 = 15.6%. It overstates by 0.7 pp due to volatility drag.

Harmonic Mean — CFA LM1: Dollar-Cost Averaging Example

$X^H = n / \Sigma(1/X_i)$ Use when averaging prices or ratios (P/E) applied to a fixed currency amount

CFA LM1 PRACTICE PROBLEM 4 — directly from CFA Vol 1

A portfolio manager invests EUR 5,000 each year for 4 years at the following prices: Year 1 = EUR 62, Year 2 = EUR 76, Year 3 = EUR 84, Year 4 = EUR 90.

Question: What is the average price paid per share?

Harmonic mean = 4 / [(1/62) + (1/76) + (1/84) + (1/90)]
= 4 / [0.01613 + 0.01316 + 0.01190 + 0.01111]
= 4 / 0.05230 = EUR 76.48

Arithmetic mean would give: (62+76+84+90)/4 = EUR 78.00 — this OVERSTATES the average cost.
Harmonic mean is lower because more shares are bought when prices are low.

Rule: Harmonic mean for prices applied to a fixed investment amount. Arithmetic mean for equal share quantities.

Key relationship

Arithmetic mean ≥ Geometric mean ≥ Harmonic mean (they are equal only if all values identical)

Money-Weighted vs Time-Weighted Return (CFA LM1)

The same portfolio can show different returns depending on WHICH measure you use. CFA requires you to know both and when each is appropriate.

Feature	Money-Weighted Return (MWR)	Time-Weighted Return (TWR)
Formula	IRR of all cash flows in and out	Geometric mean of sub-period returns
Accounts for timing?	YES — larger investments at bad times hurt MWR	NO — eliminates timing effect
Cash flow sensitivity	Sensitive — big inflow before crash lowers MWR	Not sensitive to investor cash flows
Best for measuring	Investor performance (fund client)	Manager skill (portfolio manager)
CFA preference	Investor perspective	PREFERRED by CFA for evaluating managers
When MWR < TWR	Large investment made just BEFORE a bad period	N/A
When MWR > TWR	Large investment made just BEFORE a good period	N/A

CFA Rule: Time-weighted return is the PREFERRED measure for evaluating investment managers because it removes the effect of client cash flows, which are outside the manager's control.

SECTION 03

Future Value & Compounding

Future Value — The Power of Compounding

$$FV = PV \times (1 + r)^n$$

Each period you earn interest on the ORIGINAL principal AND on all previously earned interest. Growth is non-linear.

Year	FV at r=5% (£1,000)	FV at r=10% (£1,000)	FV at r=15% (£1,000)
1	£1,050	£1,100	£1,150
3	£1,158	£1,331	£1,521
5	£1,276	£1,611	£2,011
10	£1,629	£2,594	£4,046
20	£2,653	£6,727	£16,367
30	£4,322	£17,449	£66,212

The 72 Rule: divide 72 by the interest rate to get approximate doubling time. At 6%: 72/6 = 12 years.

Example — Future Value Step by Step

You invest £2,500 today in a savings account paying 4% per year. How much will you have after 6 years?

1

Identify inputs

PV = £2,500 | r = 4% = 0.04 | n = 6 years

2

Write the formula

$FV = PV \times (1 + r)^n = 2{,}500 \times (1.04)^6$

3

Calculate $(1.04)^6$

$1.04 \times 1.04 \times 1.04 \times 1.04 \times 1.04 \times 1.04 = 1.2653$

4

Final answer

$FV = 2{,}500 \times 1.2653 = £3{,}163.22$

What is the £663 extra?

Simple interest would have earned: 2,500 × 0.04 × 6 = £600 exactly.

Actual interest earned: £663.22.

The extra £63.22 is INTEREST ON INTEREST — the compounding effect.

Compounding Frequency — Same Rate, Different Results

Starting with £1,000 and a NOMINAL rate of 12% p.a.: more frequent compounding → higher effective return.

$$FV = PV \times (1 + r/m)^{(m \times n)} \quad \text{where } m = \text{compounding periods per year}$$

Compounding	Periods/yr (m)	Effective Annual Rate	FV after 1 year (£1,000)
Annual	1	12.000%	£1,120.00
Semi-annual	2	12.360%	£1,123.60
Quarterly	4	12.551%	£1,125.51
Monthly	12	12.683%	£1,126.83
Daily	365	12.747%	£1,127.47
Continuous (∞)	e^0.12	12.750%	£1,127.50

Continuous compounding

FV = PV × e^(r×n). The limit as m→∞. Used in options pricing and some academic finance.
Formula: FV = 1,000 × e^0.12 = 1,000 × 1.12750 = £1,127.50

Effective Annual Rate (EAR) — The True Rate to Compare

$$EAR = (1 + r/m)^m - 1$$

EAR converts ANY nominal rate to its equivalent ANNUAL effective rate, making products with different compounding frequencies directly comparable.

EXAMPLE — Which savings account is better?
Account A: nominal rate 12% p.a., compounded MONTHLY
Account B: nominal rate 12.3% p.a., compounded ANNUALLY

Account A EAR = (1 + 0.12/12)^12 − 1 = (1.01)^12 − 1 = 12.683%
Account B EAR = 12.300% (compounded once a year, so nominal = effective)

WINNER: Account A — despite the LOWER nominal rate, monthly compounding gives 12.683% vs 12.300%.
Always compare using EAR, not the nominal stated rate.

Credit cards / mortgages

Credit cards quote monthly rates. A monthly rate of 1.5% = EAR of (1.015)^12 − 1 = 19.56% annually. Much higher than it appears.

EAR from continuous

EAR = e^r − 1. At r = 12%: EAR = e^0.12 − 1 = 12.750%. The highest possible EAR for a given nominal rate.

SECTION 04

Present Value & Discounting

Present Value — Discounting Future Cash Flows

$$PV = FV / (1 + r)^n = FV \times (1 + r)^{(-n)}$$

Discounting is the REVERSE of compounding. It answers: what is a future payment worth in today's money? Higher r or longer n → lower PV.

Receive £10,000 in...	At r = 5%	At r = 10%	At r = 15%
1 year	£9,524	£9,091	£8,696
3 years	£8,638	£7,513	£6,575
5 years	£7,835	£6,209	£4,972
10 years	£6,139	£3,855	£2,472
20 years	£3,769	£1,486	£611

The same £10,000 is worth VERY different amounts today depending on timing and risk. This is the entire logic of valuation.

Example — Present Value Choice Problem

You are offered two options. Your required return is 9% p.a. Which do you prefer?

Option A

£8,000 TODAY

PV = £8,000 (already in today's money)

Option B

£12,500 in 5 years

Need to calculate PV first...

1 **Calculate $(1.09)^5$**

(1.09)^5 = 1.5386

2 **Divide**

PV = 12,500 / 1.5386 = £8,124

CHOOSE OPTION B: PV = £8,124 > £8,000. Worth more in today's money at your 9% required return.

Cash Flow Timelines — A Critical Tool

Always draw a timeline before calculating. It prevents sign errors and missed cash flows. The golden rule: negatives are outflows, positives are inflows.

TIMELINE EXAMPLE: Invest £5,000 today, receive £2,000/year for 5 years. Is this worthwhile at r = 10%?

t=0: √£5,000 (invest) t=1: +£2,000 t=2: +£2,000 t=3: +£2,000 t=4: +£2,000 t=5: +£2,000

PV of each inflow: 2,000/1.10 + 2,000/1.21 + 2,000/1.331 + 2,000/1.464 + 2,000/1.611
= 1,818 + 1,653 + 1,503 + 1,366 + 1,242 = £7,582
NPV = 7,582 − 5,000 = +£2,582 → ACCEPT (NPV > 0)

Timeline rules: (1) Label t=0, t=1, t=2, etc. (2) Cash OUT is negative, cash IN is positive. (3) PV each cash flow at its specific discount factor. (4) Sum all PVs.

t	Cash Flow	Discount Factor	PV
0	√£5,000	1.000	√£5,000
1	+£2,000	1/1.10 = 0.909	£1,818
2–5	+£2,000 each	1/(1.10)^t	Σ = £7,582
NPV			= £7,582 − £5,000 = +£2,582

Example — PV of Uneven Cash Flows

A project produces different cash flows each year. Discount each separately at its specific factor, then sum.

Year	Cash Flow	Discount Factor (8%)	Present Value
1	£1,000	1/1.08 = 0.9259	£925.93
2	£2,500	$1/(1.08)^2 = 0.8573$	£2,143.35
3	£3,200	$1/(1.08)^3 = 0.7938$	£2,540.12
4	£1,800	$1/(1.08)^4 = 0.7350$	£1,323.06
		TOTAL PV	£6,932.46

If the initial investment is LESS than £6,932 → positive NPV → ACCEPT.
If the initial investment is MORE than £6,932 → negative NPV → REJECT.

Key technique: multiply each cash flow by its own discount factor

Year 1 factor = 1/(1+r)^1. Year 2 = 1/(1+r)^2. And so on.
Never apply a single factor to all cash flows unless they are equal (annuity) — that is the annuity formula shortcut covered in Section 06.

CFA LM2 Example 1 — Republic of India Discount Bond

Directly from CFA Vol 1 LM2. Discount bonds (zero-coupon) pay NO coupons: all return comes from buying below par.

$$PV = FV / (1 + r)^t$$

REPUBLIC OF INDIA GOVERNMENT BOND — CFA LM2, Example 1

Single principal cash flow payable in 20 years. YTM = 6.70%. Annual compounding. FV = INR 100.

Question 1: Price at issuance?
PV = 100 / (1.067)^20 = 100 / 3.6585 = INR 27.33

Question 2: Price 3 years later (17 years remaining, r unchanged)?
PV = 100 / (1.067)^17 = 100 / 3.0069 = INR 33.21
The INR 5.88 price rise = implied interest earned over 3 years.

Question 3: Bond price drops to INR 22.68. What is the implied YTM?
r = (100/22.68)^(1/20) − 1 = 7.70% (a 100bps rise reduced the price from 27.33 to 22.68)

CFA LM2 Example 2 — Hellenic Republic of Greece Coupon Bond

A coupon bond pays BOTH periodic coupons and final principal. Price = sum of all discounted cash flows. Directly from CFA Vol 1 LM2.

HELLENIC REPUBLIC OF GREECE — Annual Coupon Bond (CFA LM2, Example 2)

Issued during COVID: 2% annual coupon, 7-year maturity. At issuance YTM = 2.00%.

Question 1: Issuance price?
PV = 2/1.02 + 2/1.02^2 + 2/1.02^3 + 2/1.02^4 + 2/1.02^5 + 2/1.02^6 + 102/1.02^7
= 1.96 + 1.92 + 1.88 + 1.85 + 1.81 + 1.78 + 88.80 = EUR 100.00 (at PAR, because coupon = YTM)

Question 2: One year later, Eurozone inflation spiked. Bond trades at EUR 93.091. What is the implied YTM?
93.091 = 2/(1+r) + 2/(1+r)^2 + 2/(1+r)^3 + 2/(1+r)^4 + 102/(1+r)^5
Solve using RATE function: r = 3.532%

Key insight: Rising inflation → higher required yields → bond prices FALL. The investor suffers a capital loss.

Price-Yield Rule: price and yield ALWAYS move in OPPOSITE directions.

CFA LM2 Example 3 — Republic of India Semiannual Coupon Bond

Most bonds pay SEMIANNUAL coupons. Adjust: r/2 per period and n x 2 periods. Directly from CFA Vol 1 LM2.

REPUBLIC OF INDIA SEMIANNUAL COUPON BOND — CFA LM2, Example 3

20-year bond. Coupon rate = 6.70% p.a. semiannual. Par = INR 100.
PMT = 3.35 per period (= 6.70/2). Periodic rate = 3.35% (= 6.70%/2). N = 40 periods.

At issuance (YTM = 6.70%):
PV = 3.35/1.0335 + 3.35/1.0335^2 + ... + 103.35/1.0335^40 = INR 100.00
Coupon = YTM → bond prices at PAR.

If YTM rises immediately to 7.70% (r per period = 3.85%, n = 40):
PV = 3.35/1.0385 + 3.35/1.0385^2 + ... + 103.35/1.0385^40 = INR 89.88
In Excel: =PV(0.0385, 40, 3.35, 100, 0) = −89.88

A 100bps rise in yield → bond price falls 10.12% from par. This is INTEREST RATE RISK.

Semiannual rule: always divide annual YTM by 2 AND double the number of periods.

SECTION 05

NPV & IRR

Net Present Value — The Gold Standard of Investment Appraisal

$$NPV = \Sigma CF_t / (1+r)^t - \text{Initial Investment}$$

NPV directly measures how much value a project creates in TODAY'S money. It is the theoretically correct criterion.

NPV > 0 → ACCEPT	Project earns MORE than the required rate. It creates value — increases firm and shareholder wealth.
NPV = 0 → INDIFFERENT	Project exactly meets the required return. Shareholders are no better or worse off.
NPV < 0 → REJECT	Project earns LESS than required. It destroys value. Capital is better deployed elsewhere.

Example — Full NPV Calculation (Factory Upgrade)

A factory upgrade costs £50,000 today and generates cash inflows over 5 years. Cost of capital = 10%. Should you invest?

Period	Cash Flow	Discount Factor	PV
t=0	√£50,000	1.000	√£50,000
Year 1	+£12,000	1/1.10 = 0.9091	£10,909
Year 2	+£15,000	1/1.21 = 0.8264	£12,397
Year 3	+£18,000	1/1.331 = 0.7513	£13,524
Year 4	+£16,000	1/1.464 = 0.6830	£10,928
Year 5	+£14,000	1/1.611 = 0.6209	£8,693
		SUM of PVs	£56,451
		NPV = 56,451 − 50,000	= +£6,451

ACCEPT — NPV = +£6,451. The factory earns more than 10% on the £50,000 invested. It creates £6,451 of value in today's money.

Internal Rate of Return (IRR)

IRR ≈ 14.2%. Decision rule: IRR > hurdle rate (10%) → ACCEPT. In Excel: =IRR(−50000,12000,15000,18000,16000,14000) = 14.2%

4.75

Using the same factory upgrade (same cash flows). We try different rates until NPV reaches zero.

Discount Rate Tested	NPV Result	Interpretation
5%	+£14,716	Way above zero — IRR is much higher than 5%
10%	+£6,451	Positive — IRR still above 10%
13%	+£1,832	Positive but approaching zero
14%	+£398	Almost zero — very close to IRR
14.5%	√£213	Crossed zero — IRR is between 14% and 14.5%
15%	√£821	Negative
20%	√£5,824	Strongly negative

IRR — Strengths and Four Key Limitations

IRR is intuitive ('our return is 14%') but has four important limitations that CFA emphasises.

Reinvestment assumption

IRR assumes interim cash flows are reinvested at the IRR rate. If IRR = 25%, it assumes you can reinvest at 25% — usually unrealistic. NPV assumes reinvestment at r (the cost of capital), which is more realistic.

Multiple IRR problem

If cash flows change sign more than once (e.g. out, in, out), multiple IRRs may exist. Spreadsheets return just one — which may not be meaningful. Use NPV in all such cases.

Scale problem

IRR ignores project size. A small project at 30% IRR may generate less VALUE than a large project at 18% IRR. IRR is a rate, not a currency amount.

NPV always wins

For mutually exclusive projects, if NPV and IRR rank differently, ALWAYS follow NPV. It measures actual £ wealth created — the correct objective.

NPV vs IRR — When They Conflict (Scale Problem)

Mutually exclusive projects: choose ONE. IRR says A, NPV says B. Which do you follow? CFA is clear: NPV wins.

Project	Investment	Cash Flow (yr 1)	NPV @ 10%	IRR	Recommendation
Project A	£10,000	£13,500	+£2,273	35.0%	IRR says choose A
Project B	£50,000	£65,000	+£9,091	30.0%	NPV says choose B

FOLLOW NPV — Project B creates £6,818 MORE value in £ terms. IRR's higher % is on a much smaller base.

WHY NPV WINS

NPV measures the actual INCREASE in wealth in £ terms.
Project A: +£2,273. Project B: +£9,091.
You would need to invest £40k more but you gain £6,818 more value. Clearly worth it.

WHY IRR MISLEADS

35% on £10k = £3,500 gross return.
30% on £50k = £15,000 gross return.
IRR ignores scale. A smaller project at a higher rate can still create LESS value.

SECTION 06

Annuities, Perpetuities & DDM

Ordinary Annuity — Equal Payments at Period End

An annuity is a series of equal payments at equal intervals. The formula below avoids discounting each payment separately.

$$PV = PMT \times [\,1 - (1+r)^{(-n)}\,] / r$$

EXAMPLE — Car Loan

Borrow £20,000 at 6% p.a. compounded monthly. Repay in 48 monthly instalments.
r per period = 6%/12 = 0.5%. n = 48.
PMT = 20,000 × 0.005 / [1 − (1.005)^(−48)] = 20,000 × 0.005 / 0.2132 = £469.70/month
Total paid = 48 × 469.70 = £22,546. Total interest = £2,546.

Key applications

Mortgages, car loans, bond coupon streams, lease payments, pension payouts — any equal periodic payment over a fixed horizon.

Annuity factor

[1−(1+r)^(−n)]/r is the annuity factor. It gives the PV of £1/period for n periods. Multiply by PMT to get total PV.

CFA LM2 Example 5 — Mortgage Cash Flows

Directly from CFA Vol 1 LM2, Example 5. The definitive worked example for annuity calculations in finance.

QUESTION (CFA LM2, Example 5)

An investor seeks a fixed-rate 30-year mortgage to finance 80% of a USD 1,000,000 residential property. Annual mortgage rate = 5.25%.

1

Loan amount

PV = 80% × USD 1,000,000 = USD 800,000

2

Convert rate

r = 5.25% / 12 = 0.4375% per month

3

Convert periods

n = 30 × 12 = 360 monthly payments

PMT = 800,000 × 0.4375% / [1 − (1.004375)^(−360)] = USD 4,417.63 per month

Amortisation — How Payments Split Interest & Principal

Every monthly payment of USD 4,417.63 is IDENTICAL, but the split between interest and principal changes each month.

Month	Total Payment	Interest (0.4375% × Balance)	Principal Repaid	Remaining Balance
1	$4,417.63	$3,500.00	$917.63	$799,082.37
2	$4,417.63	$3,495.99	$921.64	$798,160.73
3	$4,417.63	$3,491.95	$925.68	$797,235.05
...	...	...	...	...
358	$4,417.63	$57.48	$4,360.15	$8,777.61
359	$4,417.63	$38.40	$4,379.23	$4,398.39
360	$4,417.63	$19.24	$4,398.39	$0.00

The Key Insight

Month 1: interest = £3,500 (79% of payment), principal = £918 (21%).
Month 360: interest = £19 (0.4%), principal = £4,398 (99%).
Total interest over 30 years = USD 4,417.63 × 360 − 800,000 = USD 789,347 — nearly equal to the original loan!

Annuity Due — Payments at Period Beginning

An annuity due shifts every payment one period earlier. Each payment is worth more (one extra period of earning).

$$PV(\text{Annuity Due}) = PV(\text{Ordinary Annuity}) \times (1 + r)$$

Ordinary Annuity

First payment at END of period 1 (t=1).
Examples: most loans, bond coupons, mortgage.
Pay once the period is over.

Annuity Due

First payment at BEGINNING of period 1 (t=0).
Examples: rent, insurance premiums, leasing contracts.
Pay before the period begins.

EXAMPLE — Lease Payments

Monthly lease of £500 for 12 months at 6% annual rate (0.5%/month).
Ordinary annuity: PV = 500 × [1 − (1.005)^(−12)] / 0.005 = £5,814.17
Annuity due: PV = 5,814.17 × 1.005 = £5,843.24
The annuity due is £29 more valuable because each payment arrives one month earlier.

Perpetuity — CFA LM2 Example 6: Shipline PLC

$$PV = PMT / r \quad (\text{as } n \to \infty, \text{ the annuity factor converges to } 1/r)$$

A perpetuity is an annuity with no end date. The formula is elegantly simple but extremely sensitive to r.

SHIPLINE PLC — CFA Vol 1, LM2, Example 6

Shipline PLC pays a regular annual dividend of GBP 1.50 per share indefinitely.
Shareholders require a 15% return.
PV = GBP 1.50 / 0.15 = GBP 10.00 per share

Sensitivity: if required return RISES to 20%: PV = 1.50 / 0.20 = GBP 7.50 (a 25% price DROP for a 5pp rate rise!)
If required return FALLS to 10%: PV = 1.50 / 0.10 = GBP 15.00 (a 50% price RISE)

Key lesson: perpetuities are EXTREMELY sensitive to changes in the discount rate.

Applications of perpetuity

Preferred shares with fixed dividend.
Consol bonds (UK historical).
Endowments designed to fund income forever.
Property yielding perpetual rent.

Price-rate intuition

PV = PMT/r. Double r → halve the price. Halve r → double the price. This is why long-duration assets are so sensitive to interest rate moves.

CFA LM2 Example 4 — KB Financial Perpetual Bond

Directly from CFA Vol 1, LM2, Example 4. Solving for IMPLIED YIELD from a market price — the reverse of pricing.

KB FINANCIAL (KOOKMIN BANK HOLDING COMPANY) — CFA LM2, Example 4

In 2020, KB Financial issued KRW 325 billion in perpetual bonds with a 3.30% quarterly coupon.
Market price = KRW 97.03 per KRW 100 par (a DISCOUNT to par).

Question: What is the implied YTM at this market price?

Step 1: Quarterly coupon = (3.30% × 100) / 4 = KRW 0.825 per period
Step 2: Solve PV = PMT / r for r:
97.03 = 0.825 / r
r = 0.825 / 97.03 = 0.850% per quarter
Step 3: Annualise: YTM = 0.850% × 4 = 3.40%

Conclusion: Bond trades at DISCOUNT (97.03 < 100) → implied YTM (3.40%) > coupon rate (3.30%).

Why trading at discount?

If investors now require a 3.40% yield but the bond only pays 3.30%, they will only buy it at a price BELOW par — hence 97.03. This is basic bond math: yield and price move oppositely.

Why perpetual bonds exist?

Issuers get equity-like financing with no maturity — no refinancing risk. Investors get higher yield. Common in banking (AT1 capital).

Growing Perpetuity — The Gordon Growth Model (1956)

$$PV = D_1 / (r - g)$$ where D_1 = next dividend, r = required return, g = dividend growth rate

If dividends grow at a constant rate g forever, the perpetuity formula gains one extra term: the growth subtracted from r.

QUICK EXAMPLE

A company just paid D_0 = £2.00. Dividends grow at 5%/yr forever. Required return r = 12%.

$D_1 = D_0 \times (1+g) = 2.00 \times 1.05 = £2.10$

$PV = D_1 / (r - g) = 2.10 / (0.12 - 0.05) = 2.10 / 0.07 = £30.00$ per share

D_1 (next dividend)	Higher dividend → higher price. Directly proportional.
r (required return)	Higher risk / higher rates → lower price. In denominator.
g (growth rate)	Faster growth → smaller denominator → higher price.
Requirement: r > g	If g ≥ r, the formula blows up (infinite value). Only works for mature, slowly growing firms.

CFA LM2 Example 7 — Shipline Two-Stage Dividend Growth

Directly from CFA Vol 1, LM2, Example 7. Many firms have a high early growth phase, then settle into steady-state growth.

SHIPLINE PLC TWO-STAGE DDM — CFA LM2, Example 7

Recall: Shipline constant dividend GBP 1.50/yr gives PV = GBP 10.00 at r = 15%.
Now assume: dividend grows at g_s = 6% for 3 years, then g_l = 2% forever. r = 15%.

STEP 1: PV of dividends in high-growth phase (Years 1–3):
t=1: $D_1 = 1.50 \times 1.06 = 1.590$. PV = 1.590/1.15 = 1.383
t=2: $D_2 = 1.50 \times 1.06^2 = 1.685$. PV = $1.685/1.15^2 = 1.274$
t=3: $D_3 = 1.50 \times 1.06^3 = 1.787$. PV = $1.787/1.15^3 = 1.175$
Sum of Stage 1 PVs = GBP 3.832

STEP 2: Terminal value at end of Year 3 (stable 2% growth from Year 4 onward):
$D_4 = 1.787 \times 1.02 = 1.822$. $P_3 = D_4 / (r - g_l) = 1.822 / (0.15 - 0.02) = 1.822/0.13$ = GBP 14.015
$PV(P_3) = 14.015 / 1.15^3 = 14.015 / 1.5209$ = GBP 9.217

STEP 3: Total PV = 3.832 + 9.217 = GBP 13.05

CFA answer: GBP 13.05 (vs GBP 10.00 constant, GBP 17.67 at permanent 6% growth — two-stage is between the extremes).

Gordon Growth Model — Sensitivity Analysis

Base case: D_1 = £3.00, r = 10%, g = 4%. PV = 3.00 / (0.10−0.04) = £50. The model is EXTREMELY sensitive to small changes in r−g.

PV = D_1 / (r − g). Base case in BOLD. Note how small changes can nearly double or halve the price.

g \ r	r = 8%	r = 9%	r = 10%	r = 11%	r = 12%
g = 3%	£60	£50	£43	£38	£33
g = 4%	£75	£60	£50	£43	£38
g = 5%	£100	£75	£60	£50	£43
g = 6%	£150	£100	£75	£60	£50

Three critical lessons

1. The r−g spread drives everything. Narrowing by 1 percentage point can increase value by 50–100%.
2. Never use g close to r — the model becomes meaninglessly sensitive. Practical limit: g < (r−2%).
3. Small forecasting errors in g or r compound dramatically. Use sensitivity analysis and ranges, never a point estimate.

PRACTICE PROBLEMS

Work through each before checking the answers on the next slide.

Q1 Republic of India discount bond (CFA LM2): par = INR 100, YTM = 6.70%, 20yr. If YTM RISES to 8.70%, what is the new price? Show calculation.

Q2 You invest £5,000 in Account A (8% p.a. monthly compounding) and Account B (8.2% p.a. annual compounding). Which has higher EAR? By how much?

Q3 A project costs £80,000 and generates: Year 1 = £25,000, Year 2 = £32,000, Year 3 = £28,000, Year 4 = £20,000. At r = 9%, calculate NPV. Accept or reject?

Q4 Fund Y (CFA LM1): annual returns over 5 years: 19.5%, −1.9%, 19.7%, 35.0%, 5.7%. Calculate the geometric mean return.

Q5 Shipline two-stage DDM (CFA LM2): D_0=GBP1.50, g_s=6% for 3yr, g_l=2% forever, r=15%. What is the implied stock price? (CFA answer: GBP 13.05)

Practice — Worked Solutions

A1

Periodic r = 8.70/2 = 4.35% for semiannual? No — this is annual compounding. PV = 100/(1.087)^20 = 100/5.226 = INR 19.13. (Annual 8.70%: factor = (1.087)^20 = 5.226. Compare with 6.70%: 100/(1.067)^20 = 100/3.659 = INR 27.33 — higher YTM = lower price.)

A2

Account A EAR = (1+0.08/12)^12 − 1 = (1.00667)^12 − 1 = 8.300%. Account B EAR = 8.200%. Account A WINS by 0.10pp despite same nominal rate — monthly compounding makes the difference.

A3

PV = 25,000/1.09 + 32,000/1.09^2 + 28,000/1.09^3 + 20,000/1.09^4 = 22,936+26,930+21,611+14,168 = £85,645. NPV = 85,645 − 80,000 = +£5,645. ACCEPT (NPV > 0).

A4

[(1.195)(0.981)(1.197)(1.350)(1.057)]^(1/5) − 1. Product = 1.195×0.981×1.197×1.350×1.057 = 1.984. 5th root = 1.984^0.2 = 1.1487. Geo mean = 14.87% ≈ 14.9%. CFA answer confirmed.

A5

Stage 1: D_1=1.590/1.15=1.383, D_2=1.685/1.3225=1.274, D_3=1.787/1.5209=1.175. Sum=3.832. Stage 2: D_4=1.787×1.02=1.822. P_3=1.822/0.13=14.015. PV(P_3)=14.015/1.5209=9.217. Total = 3.832+9.217 = GBP 13.05.

Common Mistakes — Do Not Fall Into These Traps

✗ **Wrong compounding period**

Annual rate with monthly payments: MUST convert. r = annual/12, n = years × 12. Never mix annual rates with monthly periods.

✗ **Nominal vs EAR confusion**

12% compounded monthly ≠ 12% effective. Always convert to EAR before comparing savings accounts, mortgages or credit cards.

✗ **Using arithmetic mean for multi-period**

Arithmetic mean overstates historical compound growth. Use geometric mean for historical, arithmetic for forward-looking expectations.

✗ **Gordon Growth with g ≥ r**

If growth ≥ required return, the formula returns negative or infinite value. Only valid when g is STRICTLY below r (at least 2pp gap).

✗ **Forgetting the t=0 cash flow in NPV**

NPV = PV of all cash flows INCLUDING the initial outflow (negative at t=0). Forgetting it gives a PV sum, not an NPV.

✗ **Choosing IRR over NPV (mutually exclusive)**

If IRR and NPV conflict for mutually exclusive projects, ALWAYS follow NPV. It measures actual value created in currency terms.

Formula Reference Card — Session 1

All formulas from this session in one place. Keep this for revision.

Holding Period Return	HPR = (End Price − Begin Price + Income) / Begin Price
Geometric Mean	$R^g = [(1+R_1)(1+R_2)\ldots(1+R_n)]^{(1/n)} - 1$
Harmonic Mean	$X^H = n / \Sigma(1/X_i)$ — use for cost-averaging & P/E ratios
Future Value	$FV = PV \times (1+r)^n$
FV (m compoundings)	$FV = PV \times (1 + r/m)^{(m \times n)}$
Effective Annual Rate	$EAR = (1 + r/m)^m - 1$
Present Value	$PV = FV / (1+r)^n$
Net Present Value	$NPV = \Sigma\, CF_t/(1+r)^t - C_0$
Ordinary Annuity PV	$PV = PMT \times [1-(1+r)^{(-n)}] / r$
Annuity Due PV	$PV = PMT \times [1-(1+r)^{(-n)}] / r \times (1+r)$
Perpetuity PV	$PV = PMT / r$
Gordon Growth Model	$PV = D_1 / (r - g)$ requires $r > g$
Nominal risk-free rate	= Real rate + Inflation premium (Fisher approx)

KEY TAKEAWAYS

Session 1 — Time Value of Money

- r = Required return = Discount rate = Opportunity cost. Decompose into 5 risk premiums.
- Geo mean (CFA: Fund Y = 14.9%) for compound history. Harmonic mean for DCA prices (CFA: EUR 76.48).
- Time-weighted return preferred for evaluating managers (removes cash flow timing effects).
- India bond (CFA LM2): PV = 100/(1.067)^20 = INR 27.33. Price and yield always move opposite.
- Greece bond (CFA LM2): priced at 93.091 after inflation spike → implied YTM 3.532% (up from 2%).
- NPV always beats IRR for mutually exclusive decisions. NPV measures actual £ value created.